

Using High-Wind Direct Covariance Air-Sea Heat Flux Observations to Understand Sea Spray Physics and Inform Tropical Cyclone Model Development for Current and Future Climates

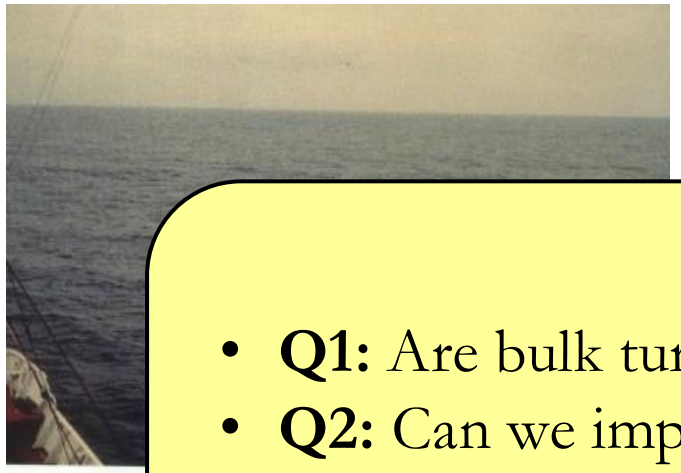
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Air-Sea Physical Processes Control Heat and Momentum Fluxes

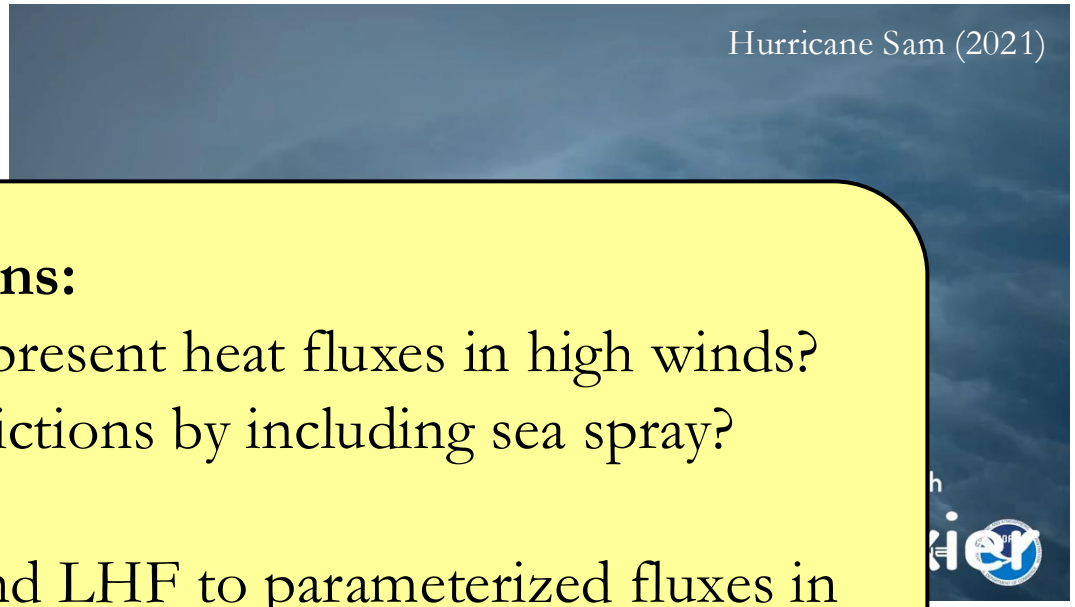
Low Winds



Moderate Winds



High Winds



Science Questions:

- **Q1:** Are bulk turbulent physics sufficient to represent heat fluxes in high winds?
- **Q2:** Can we improve high-wind heat flux predictions by including sea spray?

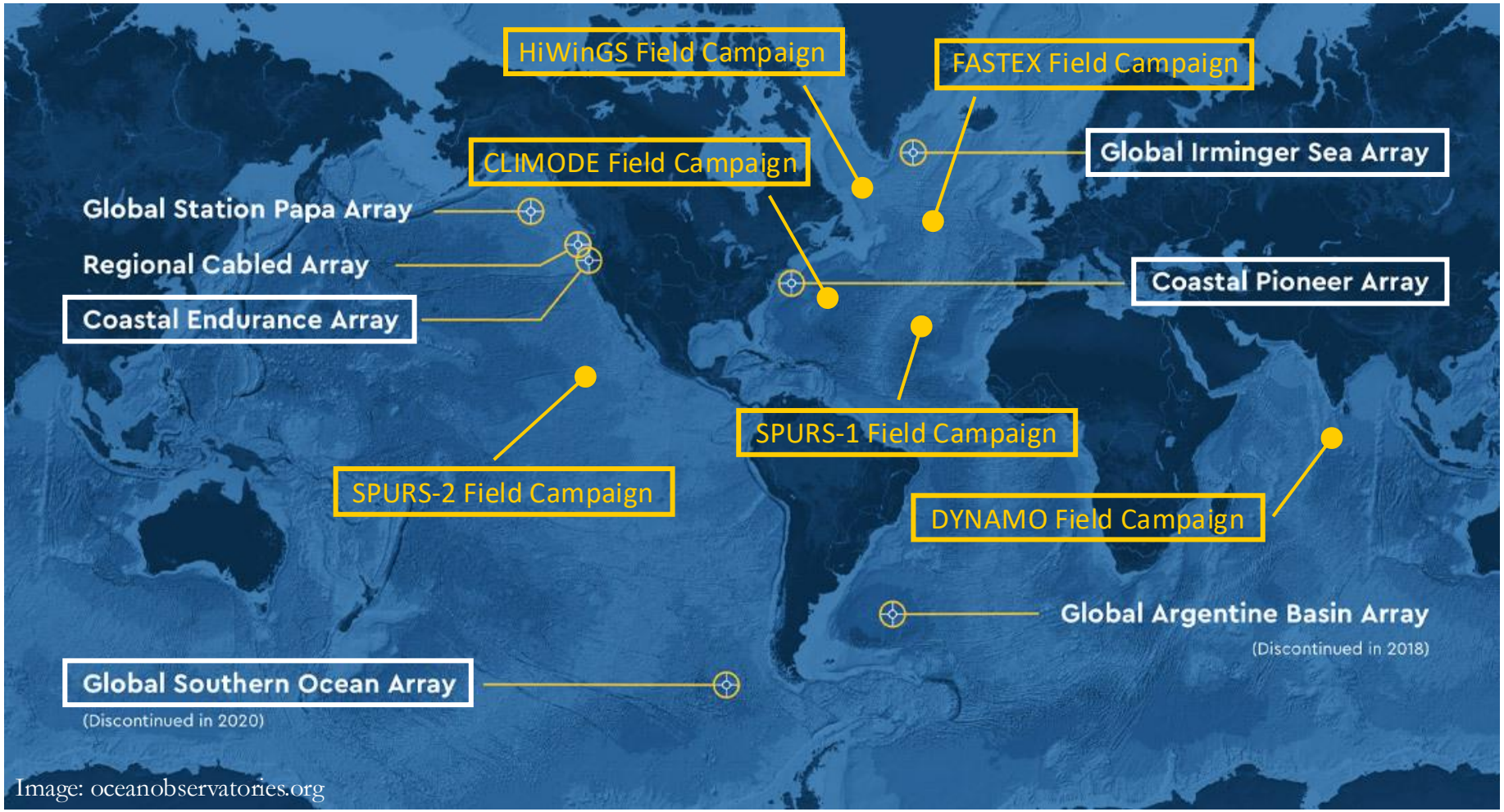
- **Approach:** Compare direct covariance SHF and LHF to parameterized fluxes in bulk-only (i.e., COARE) and bulk-spray frameworks.

$$\text{Wind Stress} = \rho \overline{u'w'} = \rho U_{10} \left[\frac{\kappa}{\ln(10/z_0) - \psi_M}\right] = \rho U_{10} C_{d10}$$

$$\text{SHF} = \rho c_p \overline{\theta' w'} = \rho c_p U_{10}(\theta_s - \theta_{10}) \left[\frac{\kappa}{\ln(10/z_0) - \psi_M}\right]\left[\frac{\kappa}{\ln(10/z_{0t}) - \psi_H}\right] = \rho c_p U_{10}(\theta_s - \theta_{10}) C_{h10}$$

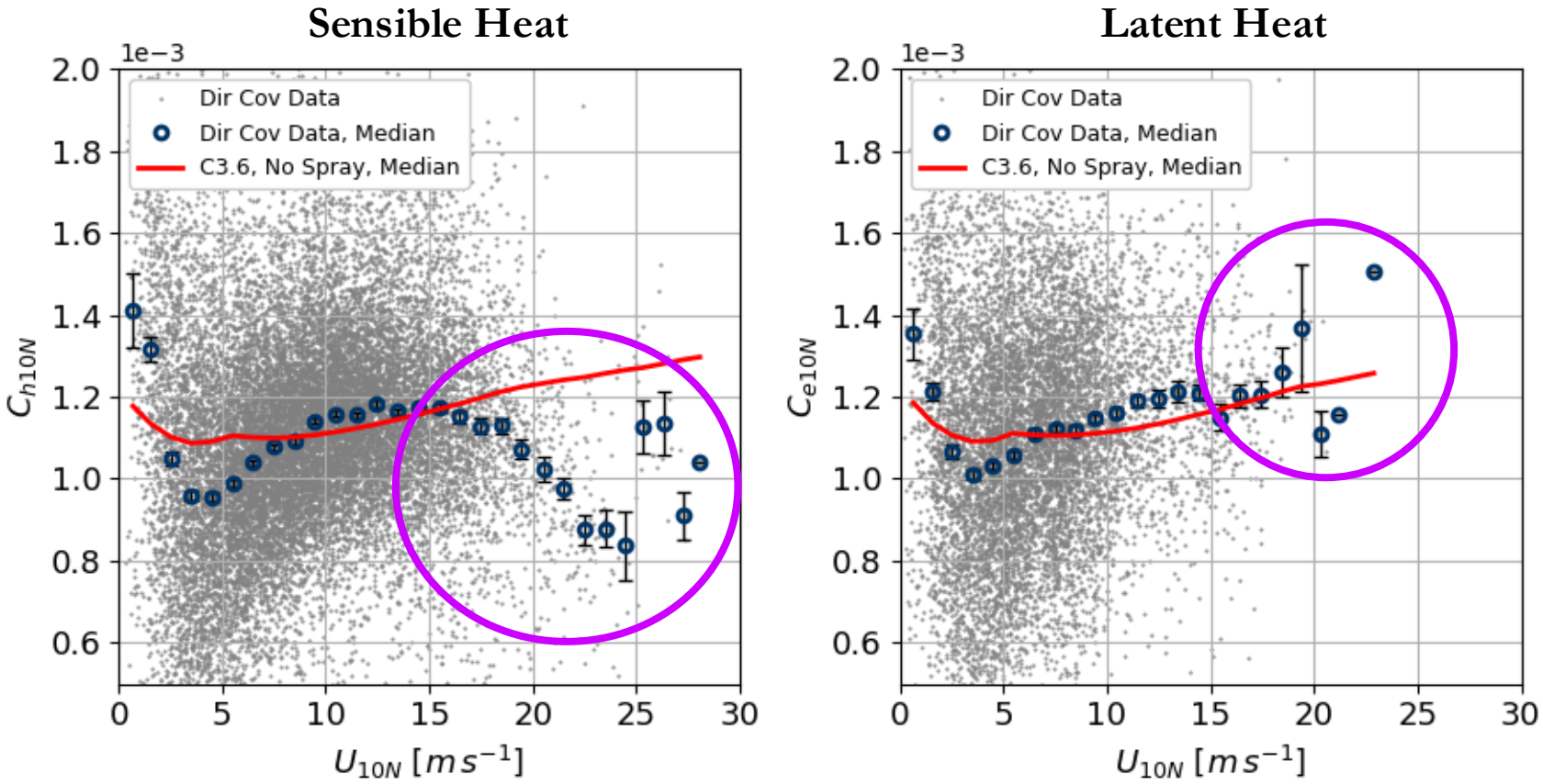
$$\text{LHF} = \rho L_v \overline{q' w'} = \rho L_v U_{10}(q_s - q_{10}) \left[\frac{\kappa}{\ln(10/z_0) - \psi_M}\right]\left[\frac{\kappa}{\ln(10/z_{0q}) - \psi_H}\right] = \rho L_v U_{10}(q_s - q_{10}) C_{e10}$$

Direct Covariance Datasets

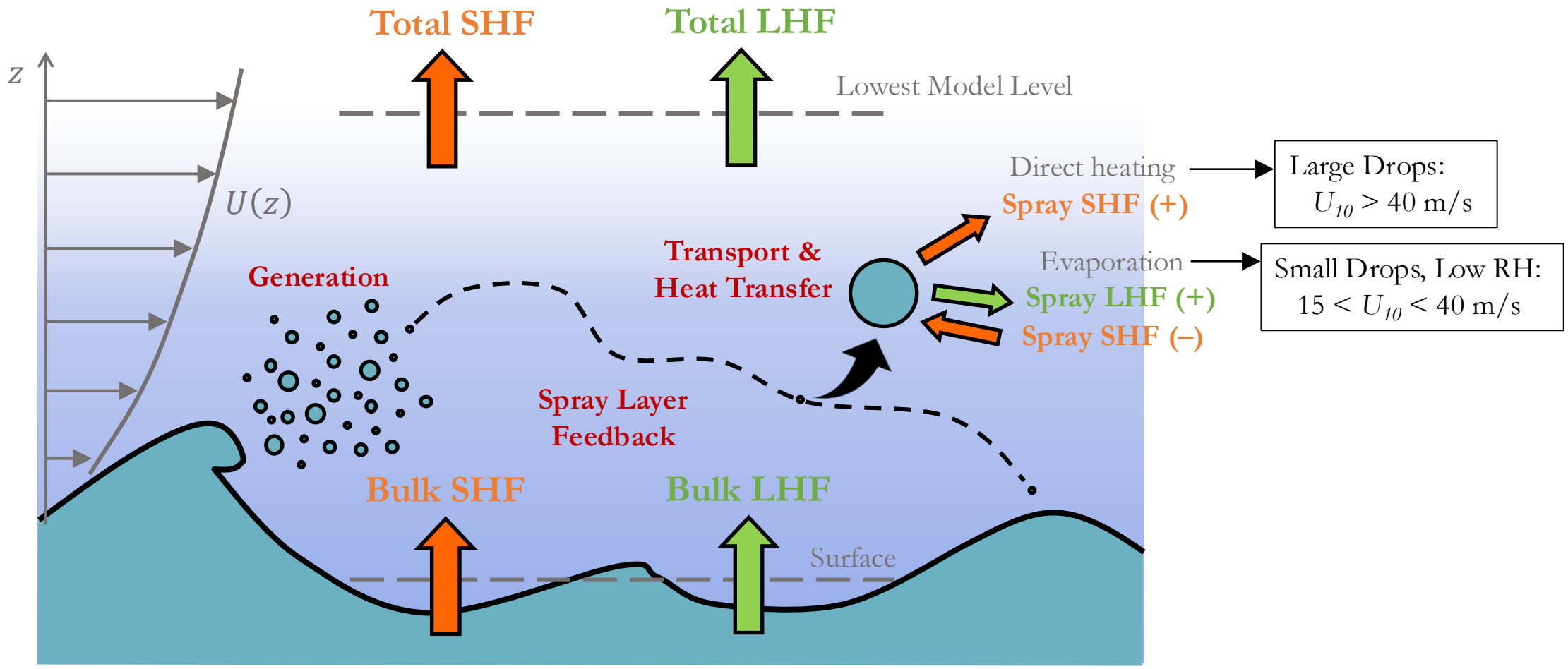


- **Sonic Anemometer**
  - Momentum flux
  - Sonic buoyancy flux (~sensible heat)
- **Infrared Gas Analyzer**
  - Latent heat flux
- **20 min averages, hourly**
- **"Good Data" Values**
  - Mom flux: 52,000
  - SHF: 19,000
  - LHF: 9,000 – and very few for $U_{10} > 20$ m/s!

Sensible and Latent Heat Show Divergent Behavior in High Winds



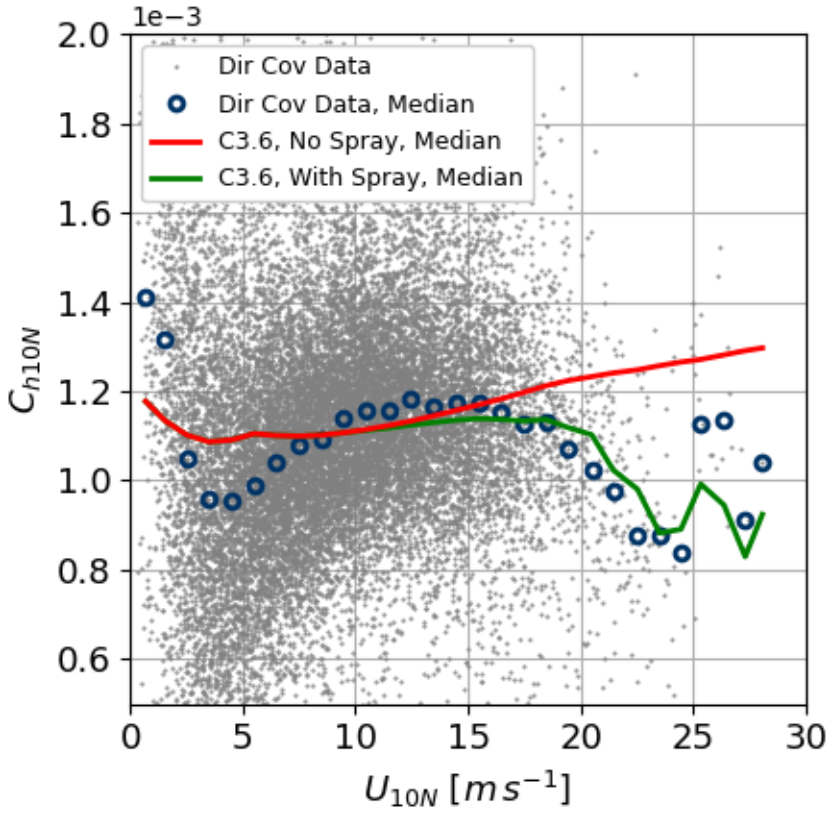
Surface Heat Fluxes with Spray



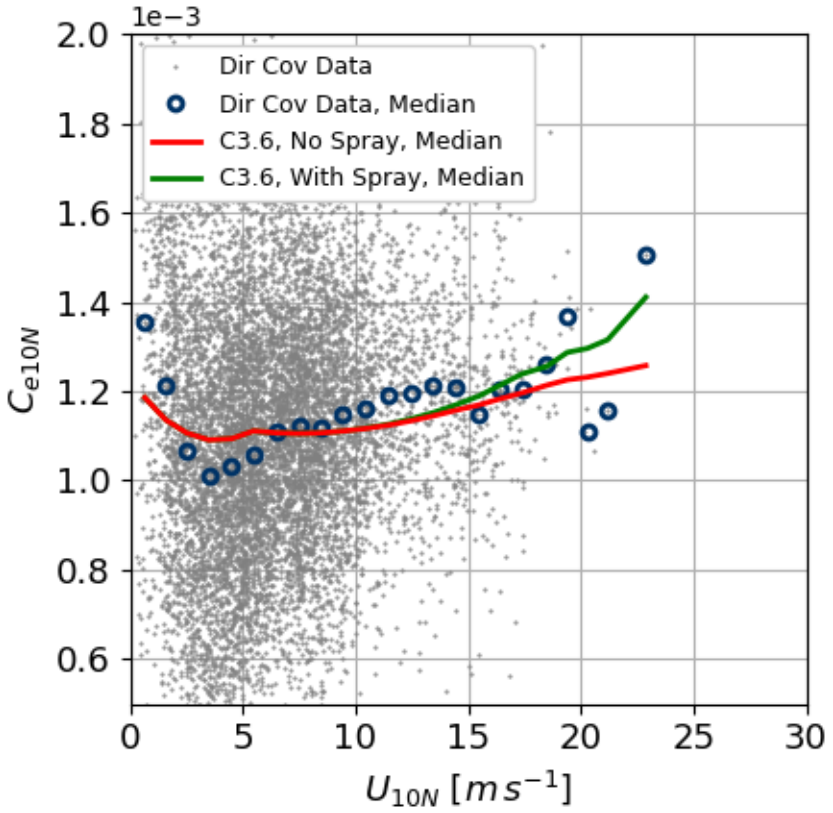
Model derivation in: Barr, Chen, and Fairall (2023), JAS.

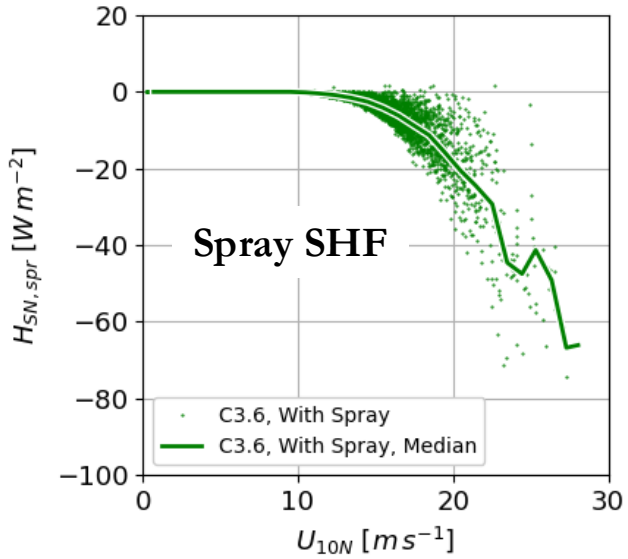
Spray Improves Prediction of Sensible and Latent Heat Fluxes

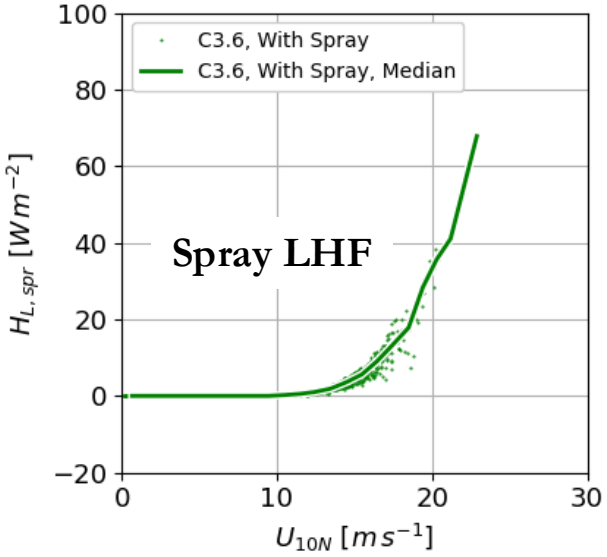
Sensible Heat



Latent Heat

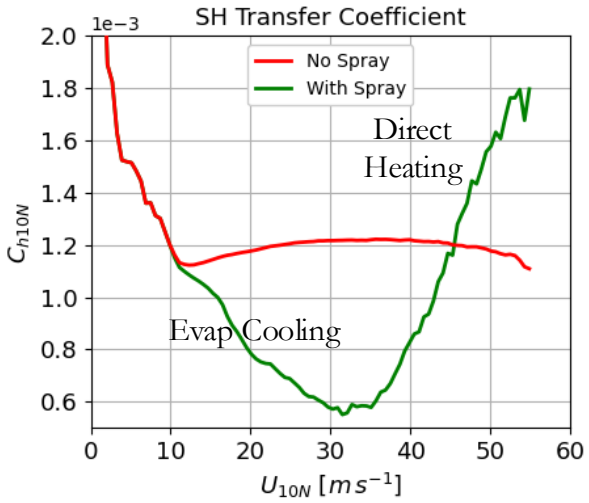


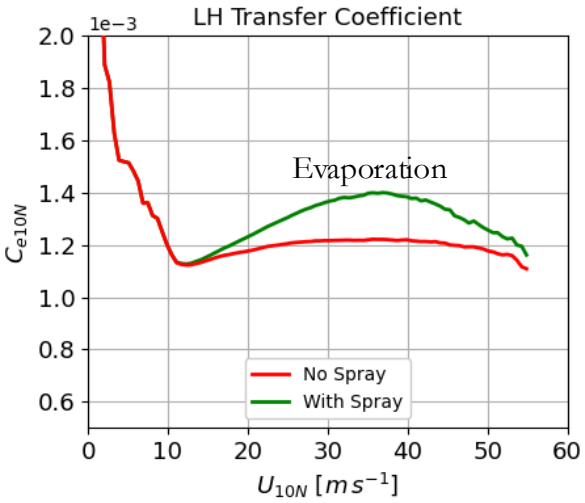




Need to Constrain Spray Heat Fluxes for Current and Future Climates!

- Testing new bulk-spray model calibration on coupled TC model output (Hurricane Michael) shows large impact of spray in extreme winds.
- Spray increases LHF by 10-15%, changes SHF by factors approaching ½ and 2.
- We must understand and model spray correctly to improve heat fluxes for current hurricane forecasts.
- In future climates with different surface thermodynamics, the contributions of bulk and spray to total heat fluxes may change, affecting TC structure and intensity.





Conclusions

- DC observations suggest that bulk turbulent physics are not sufficient for representing heat fluxes in high winds (i.e., $U_{10} > 20$ m/s).
- By incorporating spray heat fluxes into a bulk algorithm, we find improvement in heat flux predictions.
- More observations of high-wind environments, surface fluxes, and spray are needed to constrain flux parameterizations in hurricane winds.
- Bulk and spray heat fluxes may respond differently to climate change, contributing uncertainty to future TC dynamics and evolution. We must understand and constrain physical processes to improve predictive skill in future climates.

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